Fewer Clicks, Not Fewer Searches: Generative AI and the Monetization of Search

Abstract. Generative AI is widely expected to displace web search. We argue that the more immediate threat to search platforms is subtler: AI-generated answers resolve queries without the click on which sponsored search is priced. Using Google Trends data for U.S. states and Wikipedia pageviews matched to Eurostat's official statistics on generative AI use, we find little evidence that AI adoption has reduced aggregate search activity, even though individual chatbot adopters search less. Exploiting the staggered international rollout of Google's AI Overviews, we find suggestive evidence that AI summaries reduce clicks to external sites. We then develop a model in which an incumbent search platform competes with an AI-native entrant whose answers are costly to produce. The model's most robust implication is that ad-free answers are not sustainable; it also shows how inference costs are shared between users and advertisers and why a rival's adoption of advertising need not benefit the incumbent. A cannibalization channel in the model predicts that firms with larger legacy search businesses should delay full AI answers. We find no support for this prediction: across 17 search engines, and across 47 countries within Google, the timing of deployment reflects technical capability, market scale, and regulation rather than legacy revenue. A widely publicized trust shock in May 2024 was not followed by more clicks, which suggests that convenience rather than trust draws users away from search results. These findings recast the economics of AI search as a problem of monetizing answers rather than retaining searches.

Keywords: generative AI, search engines, click-through, sponsored search, AI Overviews, advertising, natural experiment, two-sided platforms, inference cost

1. Introduction

On October 29, 2026, Alphabet's shares fell by roughly 13%, their largest single-day decline since the company's 2004 listing, erasing about $475 billion in market value (Yahoo Finance 2026). The proximate cause was a quarterly report showing that revenue from Google Search had grown by 12%, a marked deceleration from 19% and 17% in the two preceding quarters, even though query volume had grown at its fastest pace in five years (Alphabet 2026a). Management attributed part of the slowdown to the "lower monetization" of AI Mode, Google's conversational answer interface. Users, in other words, were searching more, but their searches were earning less.

It is often argued that generative AI will make search engines obsolete. The episode above points to a different and more immediate threat. For two decades, search engines have monetized *intent*: a user reveals a need through a query, and the platform sells the adjacent attention to advertisers, who pay when the user clicks. Generative AI replaces a ranked list of links with a synthesized answer. When that answer satisfies the user, the click on which sponsored search depends may never occur, even though the search itself does, and each answer consumes inference compute that a list of links does not. The danger therefore lies less in fewer searches than in fewer clicks, and hence less revenue, per search.

This paper examines that transition both empirically and theoretically. We address four questions:

1. Does the adoption of AI reduce the use of search, or mainly the clicks that follow a search?
2. Can AI answers remain ad-free in equilibrium, and who ultimately bears their inference cost?
3. Do incumbents with larger legacy search businesses delay full AI answers, as theories of cannibalization would predict?
4. Do users stop clicking because they trust AI answers, or because reading search results is costly?

We begin with the premise. Across 51 U.S. states and 24 European Wikipedia editions matched to Eurostat statistics, places with higher AI adoption did not experience larger declines in search-related activity. The staggered rollout of Google's AI Overviews, by contrast, offers suggestive evidence that AI answers reduce clicks: desktop Wikipedia views fell by roughly 9% after launch, although the estimate lies at the margin of statistical significance. Together with evidence that users click a result about half as often when an AI summary is present (Chapekis 2025), this suggests that what has changed most is the number of clicks per search, the margin on which our model operates.

We then develop a model in which an incumbent with a legacy search business competes with an AI-native entrant for queries that users would rather have answered directly, and each answer carries an inference cost. Platforms choose the ad intensity of their answers, and extensions add an advertiser side, a choice between summaries and full answers, a fixed cost of answer technology, and user trust. Some results hold under weak conditions: ad-free answers are never a best response (Proposition 1); inference costs are passed on to users as additional advertising, fully when the ad price is fixed and partially otherwise (Propositions 2 and 5); a rival's advertising helps the incumbent by making the rival less attractive but hurts it by depressing the ad price (Propositions 3 and 6); and firms with scale and in-house models deploy full answers first (Proposition 9). Others rest on cannibalization: advertising restores the incumbent's profit only above a threshold of ad productivity (Propositions 4 and 7), and an incumbent with more legacy profit keeps summaries longer before switching to full answers (Proposition 8). Finally,

users tend to overtrust AI answers and can settle into a complacent or a vigilant equilibrium (Proposition 10).

The evidence supports the first set of results but not the cannibalization channel. Alphabet's market value fell on news that AI was displacing search while Meta's did not, and since 2023 Alphabet's cost-per-click growth has moved almost one-for-one against its paid-click growth, as a scarcity dividend would imply. Legacy profit, however, does not predict when full answers arrive: across 17 search engines, those with larger legacy search revenue did not move later, and within Google, AI Mode arrived later where regulators intervened and earlier in larger markets, not later where queries are worth more to advertisers. The AI Overviews errors of May 2024, a shock to trust that left the effort of reading results unchanged, were not followed by a rise in clicks, which suggests that convenience rather than trust keeps users from reading search results.

The paper contributes to three streams of IS research. For sponsored search, we show that generative answers shift the relevant margin from searches to clicks per search, with early quasi-experimental evidence. For competition between ad-supported and paid platforms (e.g., Casadesus-Masanell and Zhu 2010), we add a marginal cost per interaction and an advertiser side, which together determine who bears the cost of AI. For IT-enabled disruption, we formalize the innovator's dilemma for search and test its central prediction, which fails: capability, scale, and regulation, not legacy profit, govern how quickly incumbents respond to generative AI.

2. Related Literature

Generative AI and information search. Chatbots complete many search tasks about as accurately as a search engine and in less time, although users check facts less carefully (Xu, Feng, and Chen

2023) and learn less deeply (Melumad and Yun 2025). In the field, chatbot adopters ran more than 20% fewer traditional searches while search-ad exposure held steady (Padilla, Lam, Lambrecht, and Hollenbeck 2026), and destination sites lost activity where AI can stand in for their content: Stack Overflow lost about a quarter of its activity (del Rio-Chanona, Laurentsyeva, and Wachs 2024; Burtch, Lee, and Chen 2024) and Wikipedia articles that overlap with ChatGPT's knowledge lost views and edits (Lyu et al. 2025). The clearest effects come from AI summaries inside the search page: hiding Google's AI Overviews raised outbound organic clicks per search from 0.37 to 0.62 but did not affect sponsored clicks (Agarwal and Sen 2026), and the staggered rollout of the same feature reduced search referrals to English Wikipedia by about 5% (Khosravi and Yoganarasimhan 2026). Online Appendix C tabulates these and related studies (Table A1).

Sponsored search and advertising inside generated content. Research on search advertising has centered on position auctions (Edelman et al. 2007; Varian 2007) and on how rank and query intent shape click-through (Ghose and Yang 2009; Agarwal et al. 2011), taking the ranked results page for granted. A newer literature studies how ads could be sold within LLM output (Feizi et al. 2023; Duetting et al. 2024; Dubey et al. 2024; Hajiaghayi et al. 2024; Balseiro et al. 2025; Aggarwal et al. 2024). These studies take as given that the platform will sell ads. We ask whether ads should appear in AI answers at all, in what quantity, and under which competitive conditions.

Two-sided platforms and ad-funded versus paid competition. Platforms balance participation on each side (Rochet and Tirole 2003; Parker and Van Alstyne 2005; Armstrong 2006), and advertising is a nuisance that viewers pay for with attention (Anderson and Coate 2005). We add a marginal cost per interaction. Prasad et al. (2003) ask when a media firm should combine

subscriptions with advertising, and Casadesus-Masanell and Zhu (2010) how an incumbent should respond to an ad-sponsored entrant; our setting mirrors the latter, because the entrant may stay ad-free. For generative engines, Zhang, Jiao, Li, and Xiong (2026) derive a cutoff rule under which an ad is shown only when its immediate payoff exceeds the long-run value of a clean answer, and Wang et al. (2025) study revenue-model choice. Our model adds an advertiser side, which determines who pays for AI answers and why a rival's ads need not help the incumbent.

IT-enabled disruption. Christensen (1997) describes how incumbents underinvest in technologies that would cannibalize a profitable core. We derive when advertising in AI answers resolves this dilemma for search and test the implied prediction that firms with more to lose move later. It does not hold. Relative to prior work, we also follow AI's effects through to the search platform's own revenue and extend the rollout analysis of Khosravi and Yoganarasimhan (2026) to 25 editions and several cohorts.

3. Model

Queries. A unit mass of queries arrives each period. A fraction $s \in [0, 1]$ is *AI-amenable*: users prefer a synthesized answer to a list of links. We treat s as exogenous and increasing over time as model quality improves. The remaining $1 - s$ queries stay in legacy search, where the incumbent is the only platform and earns profit $R > 0$ per query. Before AI answers existed, the incumbent served all queries and earned R per query.

Platforms. An incumbent I (a search platform that also offers AI answers) and an entrant E (an AI-native answer engine) compete for AI-amenable queries. Each platform $j \in \{I, E\}$ chooses an ad intensity $a_j \geq 0$ for its AI answers. An answer with ad intensity a_j earns advertising revenue

$m_j a_j$, where $m_j > 0$ is ad productivity: how much advertisers pay per unit of ad exposure. It reflects targeting precision, advertiser reach, and auction depth. Every answer costs $c > 0$ in inference compute, regardless of platform. Platform j's margin per AI answer is $m_j a_j - c$.

Consumers. Users of AI-amenable queries are uniformly distributed on [0, 1], with I located at 0 and E at 1. A user at x obtains

where v is the gross value of an answer, $\gamma > 0$ is ad nuisance (the disutility per unit of ad intensity), and $t > 0$ captures horizontal differentiation, such as ecosystem integration, default placement, brand trust, or answer style. We assume v is large enough that the market is covered. The indifferent user is

Payoffs. The incumbent's profit and the entrant's profit are

Platforms choose ad intensities simultaneously. We focus on interior pure-strategy Nash equilibria ($0 < \hat{x} < 1$). The term $(1 - s)R$ is unaffected by a_I, so the incumbent's ad choice maximizes its AI profit. The legacy business matters for the incumbent's overall trajectory as s rises, which we analyze in Proposition 4.

Discussion of assumptions. Ad intensity enters revenue and consumer utility linearly, as in Anderson and Coate (2005), capturing the trade-off between answer quality and commercial content. The cost c is what distinguishes AI answers from earlier information goods, whose

marginal cost is close to zero. Allowing $m_I \neq m_E$ captures the incumbent's advantage in advertiser relationships and targeting data.

4. Equilibrium Analysis

This section characterizes ad intensities and profits. All proofs are in Online Appendix A.

4.1 Ad-Free AI Is Not Sustainable

Proposition 1 (No ad-free best response). For any rival ad intensity, $a_j = 0$ is never a best response for platform $j \in \{I, E\}$. A platform that offers ad-free AI answers earns AI profit $-s \cdot c \cdot$(its market share) < 0, and a small increase in ad intensity strictly increases its profit.

A list of links costs almost nothing to serve, but an AI answer has a positive marginal cost, so an ad-free answer is a loss on every query, and in a free-to-use service advertising is the instrument available. An incumbent that launches AI answers to retain users, without monetizing them, converts profitable legacy queries into loss-making ones.

4.2 Equilibrium Ad Intensity and Cost Pass-Through

Proposition 2 (Equilibrium). There is a unique interior equilibrium, with

The incumbent's AI profit equals $s \cdot (2t \cdot m_I/\gamma) \cdot (\hat{x}^e)^2$. In the symmetric case $m_I = m_E = m$:

(a) Ad intensity is $a^e = t/\gamma + c/m$. It rises with differentiation t and inference cost c and falls with ad nuisance γ and ad productivity m.

(b) Inference cost is fully passed through to users. Each additional dollar of inference cost per answer raises equilibrium ad revenue per answer by exactly one dollar, and equilibrium profit per platform, $s \cdot m \cdot t/(2\gamma)$, does not depend on c.

(c) With $m_I > m_E$, the incumbent sets a lower ad intensity than the entrant ($a_I^e < a_E^e$) and serves more than half of AI-amenable queries.

The term t/γ plays the role of the Hotelling markup, in units of ad load rather than price. Part (b) is the central economic insight about generative AI: users "pay" for inference with their attention, so as answers become more compute-intensive they become more ad-heavy while platforms' margins are unchanged, and falling inference costs translate into fewer ads, not higher profit. Part (c) shows that an advantage in ad productivity works like a quality advantage: a platform that extracts more per unit of ad exposure needs fewer ads to cover its costs, offers cleaner answers, and wins share.

4.3 The Cost of an Ad-Free Rival

An AI-native entrant may commit to ad-free answers, for example to build trust or because it is financed by outside capital. Let $m_I = m_E = m$ and suppose E sets $a_E = 0$.

Proposition 3 (Ad-free rival). If E commits to $a_E = 0$ and $m \cdot t > \gamma \cdot c$, the incumbent's best response is $a_I = t/(2\gamma) + c/(2m)$, its share of AI-amenable queries falls to $1/4 - \gamma c/(4tm)$, and its AI profit is

That is, at most one quarter of its profit in the symmetric ad-supported equilibrium. The entrant loses $s \cdot c$ per user it serves and requires outside funding to sustain the commitment.

Proposition 3 implies that an ad-free rival is far more damaging to the incumbent than an ad-supported one, cutting its AI profit by at least 75%, and that ad-free AI is a subsidized strategy for the entrant whose cost grows with its own success. Ad-free commitments by AI-native entrants should therefore be transitory.

4.4 When Does Advertising Offset Search Cannibalization?

Proposition 4 (Recovery threshold). The incumbent's total profit changes with AI adoption at rate

(i) If AI answers are ad-free, the incumbent's profit falls strictly with s, and becomes negative as s approaches 1. (ii) In the symmetric ad-supported equilibrium, the incumbent's profit is non-decreasing in s if and only if

Advertising in AI answers is therefore necessary but not sufficient to reverse the decline. The threshold rises with legacy profitability R and ad nuisance γ and falls with differentiation t. The incumbent's levers are to raise m by exploiting the richer intent in conversational queries, lower γ by keeping ads clearly separated from the answer, and raise t through ecosystem integration and defaults. The factor of 2 reflects competition: the incumbent captures only half of AI-amenable queries, so its AI margin per query served must be twice the legacy profit per query.

4.5 Extension: The Advertiser Side

The base model fixes the price advertisers pay per unit of exposure. We now make it endogenous, because new ad inventory from one platform lowers the price all platforms can charge.

Setup. Advertisers multi-home and treat exposures on search and AI platforms as substitutes; consumers single-home. Let ā be the number of ad units on a legacy results page. Total ad inventory per query is

and the market-clearing price per unit of exposure is $m = \mu - \eta A$, where $\mu > 0$ is the level of advertiser demand and $\eta \geq 0$ measures *advertiser-side crowding* (the base model is the case $\eta = 0$). Let $\varepsilon \equiv \eta A/m$ denote the inverse elasticity of ad demand. Each platform takes m as given. Legacy profit per query is now $R = m \cdot \bar{a}$. Profits are

We set $m_I = m_E = m$ and study the symmetric equilibrium.

Proposition 5 (Equilibrium with an advertiser side). In the symmetric equilibrium, $a = t/\gamma + c/m$, and m is the larger root of

which exists if $B^2 \geq 4\eta sc$. Inference cost is no longer fully passed through: when $\eta > 0$, a higher c raises ad intensity on both platforms, which expands inventory and lowers the ad price ($dm/dc = -\eta s/\sqrt{(B^2 - 4\eta sc)} < 0$). The incumbent loses on both businesses: AI profit $s \cdot m \cdot t/(2\gamma)$ falls, and legacy revenue $(1 - s)m\bar{a}$ falls too.

Proposition 2(b) is the special case $\eta = 0$. With crowding, users bear part of the inference cost through more ads and the advertising market bears the rest through a lower price, which also lowers the value of every legacy search ad.

Proposition 6 (A rival's ads: two opposing effects). Consider a small increase in the entrant's ad intensity, holding the incumbent's ad intensity fixed. Its effect on the incumbent's profit is

where $k = \gamma/(2t)$ and V_I is the incumbent's ad inventory per query. At the symmetric equilibrium, the effect is positive if and only if

Here w_I is the incumbent's share of total ad inventory. The rival's ads make it less attractive to consumers, so the incumbent keeps more users (the force behind Proposition 3), but they also add inventory and lower the price of all of the incumbent's ads. The second effect is larger when the incumbent owns more of the market's inventory: for a dominant platform with w_I close to 1, a rival's ads help only if ad demand is elastic ($\varepsilon < 1$), and with unit-elastic demand they leave the incumbent about where it was.

Proposition 7 (Scarcity dividend and the revised recovery threshold). In the symmetric equilibrium, the incumbent's profit is $\Pi_I = m \cdot \Lambda$, with $\Lambda \equiv (1 - s)\bar{a} + s \cdot t/(2\gamma)$, and

The first term is the base-model effect of Proposition 4 (it is non-negative if and only if $m \cdot t/(2\gamma) \geq R$). The second term is new. If an AI answer carries fewer ad units than a legacy results page ($a < \bar{a}$), the migration of queries to AI shrinks total inventory and raises the ad price, which partly offsets cannibalization. We call this the *scarcity dividend*. If AI answers carry more ads than legacy pages ($a > \bar{a}$), the term turns negative and the decline accelerates.

Advertising in AI answers remains necessary (Proposition 1), but for a dominant incumbent the profit-maximizing strategy is ad-*light* AI, since each extra ad unit in an answer also lowers the price of every legacy search ad, a loss that an AI-native entrant does not internalize.

4.6 Extension: Summaries or Answers? The Incumbent's Choice of Format

Google moved in steps: summaries above its results (AI Overviews, 2024) before a full answer interface (AI Mode, 2025). Summaries reduce organic but not sponsored clicks (Agarwal and Sen 2026), whereas AI Mode lowered monetization (Section 5). We let the incumbent choose a format for AI-amenable queries, holding $m_I = m_E = m$ and $\eta = 0$.

Formats. The entrant always serves full AI answers. The incumbent chooses among links only (L), a summary above the links (H), and a full AI answer (A). Let $\Delta > 0$ be how much users prefer a full answer to a list of links and $\Delta_O \in (0, \Delta)$ the part of that gap a summary closes, so the quality gap to a full answer is $g = \Delta$ under L, $\Delta_H \equiv \Delta - \Delta_O$ under H, and 0 under A. Under L and H the incumbent keeps its results-page ads and earns R or $R - c_O$ per query served, where $c_O < R$ is the inference cost of a summary; under A it earns $m\, a_I - c$ (Online Appendix Table A3).

Lemma. Under a format with fixed margin r and quality gap g, the entrant's best response and the incumbent's share are

The entrant shows more ads when the incumbent's format is worse. Under format A, Proposition 2 applies, with $\pi_I = s \cdot m \cdot t/(2\gamma)$.

Proposition 8 (Summaries first, answers later). (i) Summaries dominate links if and only if

which holds whenever summaries are cheap relative to the quality they add. (ii) Full answers dominate summaries if and only if the remaining quality gap exceeds a threshold:

(iii) The threshold rises with legacy profitability R and inference cost c, and falls with ad productivity in AI answers m and with the cost of summaries c_O.

As AI answers improve relative to summaries (Δ_H rises), the incumbent first adopts summaries, which defend its users without touching its ad revenue, and switches to full answers only when the remaining gap is large enough. The switch comes later the more profitable legacy search is, which is the innovator's dilemma in a precise form, and better monetization of AI answers (higher m) brings it forward.

An incumbent may even switch to full answers when doing so leaves its profit below the pre-AI level, to limit the loss of users to the entrant (Online Appendix D).

4.7 Extension: Scale, Capability, and the Timing of Full Answers

Proposition 8 ignores the fixed cost of being able to offer full answers at all (building or licensing a model, serving infrastructure, safety work), which matters because small search engines with little legacy profit did not move early (Section 5.5). We add a fixed cost and let engines differ in scale.

Setup. Search engine j handles a share $\sigma_j \in (0, 1)$ of queries in its market, and its legacy profit per query rises with its position, $R(\sigma) = R_0 + R_1\sigma$ with $R_1 > 0$ (larger engines have deeper ad auctions). It currently serves AI-amenable queries with summaries (format H). Moving to full answers costs a fixed amount K_j per period, which is lower for engines with an in-house frontier model. Answer quality improves over calendar time d, so the remaining gap $\Delta_H(d)$

rises and the incumbent's share under summaries, $\hat{x}_H(d) \equiv \hat{x}(\Delta_H(d))$ from the Lemma in Section 4.6, falls. Let $\pi_A = m\,t/(2\gamma)$ denote the profit per AI-amenable query under full answers. Engine j switches at the first date d at which

The bracket is the gain per query, which falls with legacy profit; the factor σ_j scales it by the engine's volume, against which the fixed cost is spread.

Proposition 9 (Mid-sized engines move first). Let $D(\sigma, K)$ be the switching date. (i) D is quasi-convex in σ: at any date d, the engines that have switched form an interval of shares $[\sigma_L(d), \sigma_H(d)]$, which widens over time. The first engines to switch have intermediate shares; small engines switch later because they lack scale, and large engines switch later because they lose more legacy profit. (ii) D is increasing in K, so engines with in-house models switch earlier at every share. (iii) An engine for which $\sigma[\pi_A - (R(\sigma) - c_O)\hat{x}_H(d)] < K$ at every date never switches unless K falls, for example when open or licensable models become cheaper.

Proposition 9 replaces the monotone prediction of Proposition 8 with an inverted-U relation between scale and the speed of adoption, and adds capability as a separate determinant. A numerical example is in Online Appendix D.

4.8 Extension: User Trust and the Decision to Click

A natural explanation for the fall in clicks is that users trust AI answers and see no need to check them. We model that trust explicitly, let it be learned, and ask when trust, rather than effort, explains the decline.

Setup. An AI answer is correct with probability q, its true accuracy. A user's trust θ is the accuracy the user believes the answer has. Each query carries a stake L, the user's loss from acting on a wrong answer, drawn from a distribution F across queries and users. Clicking through to sources to verify the answer costs effort $e > 0$ and reveals any error. A user clicks if and only if the expected loss avoided exceeds the effort:

where ω is the share of answers that users verify. G is decreasing: more trust, fewer clicks. Clicks also fall when effort rises or stakes are low, whatever the level of trust.

Learning. Users learn about accuracy from the errors they detect. An error is detected for sure when the user verifies, and only with probability ζ otherwise, for example when its consequences become visible. We assume users treat the absence of a detected error as evidence of a correct answer ("no news is good news"), a pattern documented as automation bias. Their long-run trust is then

Trust and clicking are jointly determined: an equilibrium is a pair (θ, ω) with $\omega = G(\theta)$ and $\theta = T(\omega)$, that is, a fixed point of $\Phi(\theta) \equiv T(G(\theta))$.

Proposition 10 (Trust traps). (i) An equilibrium exists, and in every equilibrium users overtrust AI answers: $\theta - q = (1 - q)(1 - \zeta)(1 - \omega) \geq 0$, with more overtrust the fewer users click. (ii) If Φ crosses the 45-degree line from below at some point, there are at least three equilibria: a *vigilant* one, with low trust and many clicks; a *complacent* one, with high trust and few clicks; and an unstable one between them. (iii) If AI answers are accurate enough that $e/(1 - q)$ exceeds the

largest stake, the complacent equilibrium is unique. (iv) With multiple equilibria, a temporary rise in the detection of errors, such as widely publicized AI mistakes, can move users permanently from the complacent to the vigilant equilibrium.

A numerical example in Online Appendix D shows three equilibria when $q = 0.6$ and only the complacent equilibrium when $q = 0.9$.

Trust or effort? The model separates the two explanations by the stakes of the query: low trust raises clicks only where stakes justify the effort. Three implications follow. First, clicks should fall most for low-stakes queries: the effect of AI Overviews in the field experiment of Agarwal and Sen (2026) came from informational queries, and users check AI answers more often for health and politics (Simon, Nielsen, and Fletcher 2025). Second, distrust need not imply clicking: 82% of 2025 U.S. respondents were skeptical of AI answers, but only about 8% always check the sources (Exploding Topics 2025). Third, if trust drives clicking, salient errors should raise verification clicks (Proposition 10(iv)), such as the widely shared AI Overviews errors of late May 2024 that led Google to restrict the feature (Engadget 2024). Section 5.6 tests this prediction.

Effort, not trust, as the main driver. A user can read search results, at effort e_S and with accuracy q_S, or an AI answer, at lower effort $e_A < e_S$ and perceived accuracy θ. With benefit B from a correct answer, the user chooses the AI answer when $e_S - e_A > (q_S - \theta)B$. Even a user who trusts AI answers less than search results chooses them for low-stakes questions, so a pure trust shock, which lowers θ while leaving effort unchanged, should move users back to search results only for high-stakes questions and leave aggregate clicks largely unchanged. Section 5.6 tests this implication.

5. Empirical Analysis

Section 5.1 examines the premise that people who use AI more rely less on search. Sections 5.2 to 5.4 test the monetization predictions using stock returns around AI news, quarterly segment revenue for 2019Q4 to 2026Q3, and Alphabet's paid clicks and cost per click. Section 5.5 tests the timing predictions of Propositions 8 and 9 across 17 search engines and within Google across 47 countries, Section 5.6 uses the May 2024 AI Overviews errors to ask whether trust or effort drives the fall in clicks, and Section 5.7 summarizes. Additional material is in Online Appendix E. Two findings frame what follows. The premise survives only in a narrower form: AI reduces the clicks that follow a search, but we find no evidence that it reduces the number of searches. And while news that AI is displacing search lowers Alphabet's market value without moving Meta's, the evidence on monetization is weaker, and the prediction of Proposition 3 is not supported.

5.1 First Stage: Does AI Use Reduce Search Use?

The model's premise is that queries migrate from link-based search to AI answers as AI improves (s rises). Published evidence divides by what is measured (Online Appendix Table A2). Clicks and visits to destinations that search used to supply decline: a result was clicked on 8% of visits to pages with an AI summary versus 15% without (Chapekis 2025), and Wikimedia reported an 8% fall in human pageviews in 2025 (TechCrunch 2025). Search volume is mixed: Safari searches fell in April 2025 (Fortune 2025) and chatbot adopters search less (Padilla et al. 2026), but Google's total queries are growing at their fastest pace in five years (MarketBeat 2026). We add three tests (Table 1).

Tests A and B: AI adoption across places. Test A uses monthly Google Trends series for 51 U.S. states (including the District of Columbia), January 2019 to September 2026. The outcome is the log share of question-type queries that AI can answer ("how to," "what is") relative to queries it cannot easily replace ("weather," "near me"), and treatment is a state's standardized search intensity for ChatGPT interacted with a post-December 2022 indicator. Test B uses Wikipedia pageviews, which mostly arrive from search engines, as a proxy for clicks out of search for 24 editions whose readers are concentrated in one European country (Wikimedia Foundation 2026), matched to Eurostat's official 2025 measure of generative AI use (Eurostat 2026), which ranges from 17% in Türkiye to 56% in Norway. We include location and month fixed effects, cluster standard errors by state or edition, and in one specification control for each country's 2022 internet use interacted with the post period, because AI adoption tracks general digitalization. Eurostat's measure is available only for 2025, so treatment intensity is cross-sectional (Online Appendix E).

Test C: Staggered launch of AI Overviews. Tests A and B compare places by how much they use AI, which is not randomly assigned. Google's AI Overviews, by contrast, reached different language markets at different times for reasons unrelated to local search trends: the United States in May 2024 (Google 2024a), other countries in August 2024 (Android Authority 2024), Europe in March 2025 (PPC Land 2025), more than 40 languages in May 2025 (Google 2025a), and France in July 2026 (PPC Land 2026b). An AI Overview answers the query on the results page, so it should reduce clicks to sources such as Wikipedia. Khosravi and Yoganarasimhan (2026) use a related design with English, German, and French articles and estimate a reduction of about 5% in search referrals; ours extends it to 25 editions and several cohorts and separates desktop from mobile traffic. We estimate group-time effects with the Callaway and Sant'Anna (2021)

estimator, using not-yet-treated editions as controls and bootstrapping over editions, and focus on the August 2024 cohort, the only one with a large control group (19 editions).

Table 1. AI adoption and search-related activity. Coefficient on adoption × post (standard error clustered by state or edition; p-value).

Test	Specification	Estimate	Reading
A. U.S. states (N = 4,743)	Question-type vs. other queries	+0.024 (0.004; p < 0.01)	Wrong sign; “near me” and “weather” fell in high-adoption states
B. 24 European editions, Eurostat AI use (N = 2,232)	Difference-in-differences, all platforms	−0.063 (0.041; p = 0.13)	Not significant
B	Adding daily internet use × post (N = 2,139)	−0.036 (0.066; p = 0.59)	Disappears
B	Long difference, 2021–22 to 2025–26 (N = 24)	+0.30 pp per point of AI use (p = 0.09)	Wrong sign
C. AI Overviews launch, Aug. 2024 cohort (25 editions)	All platforms, months 0 to +6	+0.022 (0.046; 95% CI −0.072 to 0.109)	No effect on total views
C	Desktop	−0.089 (0.046; 95% CI −0.177 to −0.000)	About 9% fewer desktop views
C	Mobile web	+0.062 (0.058)	None; pre-trend (+0.111)
C	Desktop, excluding Portuguese; excluding Hindi	−0.061 (0.040); −0.090 (0.054)	Same sign, not significant

Note. Online Appendix Table A4 reports all specifications.

Results. Tests A and B find no evidence that places adopting AI more intensively reduced search-related activity more. The Test A estimate has the wrong sign, and the Test B estimate is negative but not significant and shrinks further once general internet use is controlled for; editions in high-adoption countries dipped briefly in 2023, but the gap closed by 2025 and reversed in 2026. Test C gives the closest evidence for the narrow premise: desktop Wikipedia views fell by about 9% after AI Overviews launched, with no pre-trend and negative estimates in all seven months after launch (Online Appendix Figure A1). The estimate is at the edge of

significance, however, it loses significance when single editions are dropped, and total views, which are dominated by mobile, did not change.

Interpretation. The data support a narrower premise: AI reduces the clicks that follow a search, rather than the number of searches. Users still query Google, increasingly through AI Overviews and AI Mode, but click less, and Google's paid-click growth slowed to 2–4% in 2025 even as queries grew (Section 5.4). We therefore restate the premise as follows: *as AI use rises, a growing share of queries is resolved without a click on a link.* This is compatible with individual-level evidence that chatbot adopters search less (Padilla et al. 2026), which our aggregate tests may miss (Google Trends measures shares, not volumes), and it is the case that Propositions 4–7 need: AI-amenable queries stay on the incumbent's platform but are answered by AI.

5.2 Event Study

We use daily adjusted closing prices for Alphabet (GOOGL), Meta (META), and the S&P 500 ETF (SPY) from Yahoo Finance. For each event we estimate a market model on trading days [−280, −31] and compute the cumulative abnormal return (CAR) over [0, +1]. Meta, which sells digital advertising but does not depend on search, serves as a control. We classify events ex ante: *substitution* events signal that AI displaces search (Proposition 4 predicts a negative Alphabet CAR), and *monetization* events signal that advertising enters AI answers, by Google (Propositions 1–2) or by an AI-native rival (Proposition 3), and predict a positive CAR.

Table 2. Abnormal returns around AI events, CAR[0, +1] in % (t-statistic in parentheses). Newest first.

Event day	Event	Type	Sign	GOOGL	META	GOOGL − META
2026-10-29	Alphabet Q3 results; “lower monetization” of AI Mode	Substitution	−	−12.59 (−5.44)	+3.79 (1.19)	−16.38
2026-07-23	Alphabet Q2 results; capex guidance raised	Cost shock	−	−5.62 (−2.60)	−3.21 (−1.16)	−2.41
2026-02-09	ChatGPT ads go live (U.S.)	Monetization (rival)	+	−1.94 (−0.83)	+1.18 (0.50)	−3.12
2026-01-16	OpenAI announces ad test in ChatGPT	Monetization (rival)	+	−0.37 (−0.15)	−0.07 (−0.03)	−0.30
2025-05-21	Google Marketing Live: ads in AI Mode	Monetization (own)	+	+6.01 (2.86)	+2.18 (0.95)	+3.83
2025-05-07	Apple executive: Safari searches fell for the first time	Substitution	−	−7.01 (−3.34)	+0.35 (0.15)	−7.36
2024-05-21	Google Marketing Live: ads coming to AI Overviews	Monetization (own)	+	−0.26 (−0.13)	−0.45 (−0.18)	+0.18
2022-11-30	ChatGPT released	Substitution	−	+2.17 (1.30)	+4.71 (1.27)	−2.54

Results. Substitution news is priced as a search-specific shock. The two events that revealed actual displacement, falling Safari searches and weaker AI Mode monetization, produced Alphabet CARs of −7.0% and −12.6% (both $p < 0.001$), while Meta’s CARs were insignificant; across the three substitution events the pooled standardized CAR for Alphabet is $Z = -4.32$ (mean CAR −5.8%). ChatGPT’s release produced no immediate reaction. The monetization evidence is mixed: Google’s announcement of ads in AI Mode was followed by a +6.0% CAR ($t = 2.86$), but the 2024 announcement of ads in AI Overviews produced no reaction, and across the four monetization events the pooled Z is 0.88, not significant. The October 2026 decline also contains a cost component, since the July 2026 capex announcement alone produced a −5.6% CAR.

5.3 Quarterly Evidence: Search Revenue and the Advertising Cycle

We collected quarterly “Google Search & other” revenue from Alphabet’s earnings releases and Meta’s revenue (about 98% advertising, a benchmark for the advertising cycle) from XBRL

filings: 28 quarters of year-over-year growth, 2019Q4–2026Q3, 13 before ChatGPT and 15 after. Before ChatGPT, Search growth swung with the advertising cycle, apart from 2021Q3–2022Q4, when Apple's App Tracking Transparency (ATT) depressed Meta specifically. Afterwards Search rises much less than Meta in the 2023–2026 upswing (Online Appendix Figure A2). We estimate

where g^S and g^M are the year-over-year growth of Search and Meta revenue, Post = 1 from 2023Q1, k counts quarters since 2022Q4 in the post period, ATT = 1 for 2021Q3–2022Q4, and COVID = 1 for 2020Q2–2021Q2. Standard errors are Newey–West with four lags.

Table 3. Search revenue growth and the advertising cycle, 2019Q4–2026Q3 (N = 28). Coefficients with HAC standard errors in parentheses.

Variable	(1) Level shift	(2) Post-ChatGPT trend	(3) Sensitivity break
Meta growth g^M	1.109*** (0.129)	1.159*** (0.121)	1.273*** (0.117)
g^M × Post			−0.681*** (0.119)
Post	−0.651 (0.913)	2.947 (2.343)	12.549*** (2.291)
Quarters since ChatGPT, k		−0.452** (0.223)	
ATT	18.464*** (1.877)	19.049*** (1.728)	20.384*** (2.015)
COVID	−3.937 (3.416)	−4.394 (3.233)	−5.440* (2.880)
Constant	−9.395*** (2.480)	−10.352*** (2.323)	−12.538*** (2.233)
R^2	0.83	0.84	0.88

Note. *** p < 0.01, ** p < 0.05, * p < 0.10.

Taken at face value, columns (2) and (3) support the model. After ChatGPT, Search growth falls behind the advertising cycle by about 0.45 points per quarter, and its sensitivity to the cycle drops from 1.27 to 0.59. Search revenue participates in only about half of an advertising upswing.

Placebo tests do not support a causal reading. Breaks placed before ChatGPT fit as well or better; the best fit (lowest AIC) is 2021Q3, the start of the ATT shock to Meta (Online Appendix Table A5). Without the ATT and COVID controls the sensitivity break is not significant (−0.18, p = 0.19). We conclude that quarterly firm-level data are consistent with weaker Search growth since 2023 but cannot attribute it to AI.

5.4 Advertiser-Side Evidence: Paid Clicks and Cost-per-Click

Propositions 6 and 7 rest on advertiser-side crowding ($\eta > 0$): the price of a search ad should rise when inventory grows more slowly and fall when it grows faster. Alphabet reports the year-over-year change in paid clicks and cost-per-click (CPC) for Google Search & other in each Form 10-Q, which we collected for the first three quarters of 2021–2026 (N = 18). In 2024–2025, paid-click growth slowed to 2–7% while CPC growth held at 7–8%; in 2026, as AI Mode ads expanded, click growth rose to 13% and CPC growth fell to 3–5% (Online Appendix Figure A3).

Table 4. CPC growth regressed on paid-click growth, controlling for advertiser demand (Meta revenue growth). HAC standard errors (two lags) in parentheses.

Variable	(1) 2021–2026	(2) 2021–2026	(3) 2023–2026
Paid-click growth	0.133 (0.282)	−0.750*** (0.263)	−1.020*** (0.100)
Meta revenue growth	0.355*** (0.083)	0.570*** (0.064)	0.538*** (0.024)
Post-2023		−12.142*** (3.207)	
N	18	18	12
R^2	0.53	0.64	0.94

Note. *** p < 0.01.

Once the 2021–2022 period is separated out, CPC growth falls almost one-for-one with paid-click growth (−1.02 in column 3), consistent with $\eta > 0$ and an inverse demand elasticity ε close to 1. Because Google holds most search ad inventory (w_I close to 1), εw_I is close to 1 and, by

Proposition 6, the consumer-side gain from ChatGPT's ads and the advertiser-side loss roughly cancel, which matches the null result in Section 5.2. The same estimate supports the scarcity dividend of Proposition 7: when AI answers reduced click growth in 2024–2025, higher prices kept Search revenue growing at 11–14%. Two caveats apply: a coefficient near −1 is also what fixed advertiser budgets would produce mechanically, and the sample is small, with the post-2023 relationship failing when 2022 is included (−0.13, p = 0.75).

5.5 Timing of Full AI Answers: Bing, Search Engines, and Google's Markets

Bing. Proposition 8 implies that an engine with little legacy profit relative to its rival should adopt full answers early, without the summary step. Bing is such a firm relative to Google, although its absolute legacy revenue is large (search advertising revenue of $3.8 billion in January–March 2026, against $60.4 billion for Google Search & other; Microsoft 2026; Alphabet 2026c). Microsoft announced a chat-based Bing with full AI answers in February 2023 and ads inside chat answers in March 2023 (Engadget 2023), 15 and 27 months before Google's summaries and full answers. Bing's worldwide desktop share of search referrals (StatCounter 2026) also follows the prediction: its monthly trend rose from +0.10 to +0.26 percentage points after Bing Chat, stalled (+0.02) once Google added summaries, and reversed (−0.16) after AI Mode, although in the United States it rose again after AI Mode (Online Appendix Figure A4 and Table A6). Because referral shares depend on browser defaults and a segmented trend fits a single firm's history, this is consistent with Proposition 8 but not a causal test.

Cross-firm test. Proposition 8 predicts that the larger a search engine's legacy business, the later it moves to full AI answers and the more likely it is to add summaries first. We coded the first public launch of each format for 17 search engines (Online Appendix Table A7) and measured

legacy strength by each engine's 2022 share of its home market (StatCounter 2026), with timing in months from ChatGPT's release (December 2022). The prediction is not supported: legacy share and timing are not significantly related (Spearman $\rho = +0.17$, $p = 0.51$ for all 17 engines; -0.21, $p = 0.46$ excluding AI-natives; Online Appendix Table A8), and the sign depends on whether AI-native services are included. The summary step is common among large engines (three of the four with more than 20% home share added summaries first, against four of nine below 5%), but the smallest engines were not early (Seznam, Sogou, Daum, Qwant, Ecosia, and Yahoo reached full answers late or not at all), even though they had little to lose.

Capability and legacy revenue. The delays of small engines suggest that scale and capability also govern timing, as Proposition 9 predicts. Engines with in-house frontier models reached full answers about 12 months earlier (21.6 vs. 33.1 months), in line with part (ii) of Proposition 9, but the difference is not significant ($p = 0.11$), and there is no inverted-U relation with share (Online Appendix Table A9). Market share measures position, not what an engine stands to lose, so we also collected 2022 search advertising revenue for 13 engines (Online Appendix Table A10). Engines with more legacy revenue reached full answers somewhat earlier (Spearman $\rho = -0.35$, $p = 0.24$, $N = 13$). Bing, with the second-largest legacy revenue ($12.0 billion), moved first, Google took 29 months, and Naver 42; Scale and capability thus appear to govern timing more than legacy size, supporting the fixed-cost channel of Proposition 9 but not the cannibalization channel of Propositions 8 and 9.

Within-firm test: Google across countries. Holding the firm fixed removes the confound between legacy profit and capability. We dated the first month in which AI Overviews and AI Mode were available in each of 49 countries (Google Search Help 2026) and regress months

until availability on log GDP per capita (a proxy for legacy profit per query), log internet users (a proxy for scale), and indicators for the EU/EEA, English-speaking markets, and the two markets held back by regulatory disputes, France and Türkiye (N = 47; World Bank 2026; Online Appendix Table A11). Regulation dominates: the EU/EEA received AI Mode about two months later than comparable markets, and France and Türkiye seven to eight months later still. Scale speeds adoption: larger markets received AI Mode earlier, in any language outside the EU ($p = 0.04$) and in the local language ($p < 0.01$), as the fixed-cost channel of Proposition 9 predicts. And there is no sign of a cannibalization delay: markets where a query is worth more to advertisers did not receive full answers later, and in their local languages received them earlier ($p < 0.01$). Rollouts in waves leave little independent variation, so these estimates describe Google's sequencing rather than causal effects.

5.6 A Trust Shock: The AI Overviews Errors of May 2024

The widely publicized AI Overviews errors of late May 2024 lowered trust without changing the effort of reading results, so they test Section 4.8. If trust drives clicking, clicks out of search should rise after the errors (Proposition 10(iv)); if effort drives it, they should change little. Using daily pageviews of English Wikipedia, the edition most exposed to AI Overviews in the United States (then the only market), we compare English with the ten other largest editions in a triple difference (English minus other editions, 2024 minus the same weekdays of 2023, relative to the four weeks before the May 14 launch), with Newey–West standard errors and a placebo in which each other edition is treated in turn (Online Appendix Table A12 and Figure A5). Clicks did not rise after the trust shock. Views did not change in the week after launch (desktop −0.8%, $p = 0.74$) and fell further relative to other editions after the errors went viral, by about 4%

through June and July (desktop −4.9% in June, $p < 0.01$), even though Google cut back AI Overviews from about 11% to 7% of queries by the end of June (Search Engine Land 2024), which on its own should have raised clicks. This is the opposite of what the trust account predicts and is consistent with effort as the main driver. The evidence is weak, however: the declines are not significant relative to other large editions (placebo $p = 0.40$ to 0.70), and English Wikipedia is an imperfect measure of U.S. clicks. With these caveats, convenience, more than trust, appears to keep users from reading search results.

5.7 Summary of Evidence and Remaining Tests

Table 5. Model predictions and empirical support.

Prediction (source)	Evidence	Verdict
Premise, narrow form: AI reduces clicks after a search, not (in aggregate) searches (model setup)	Click-through 8% vs. 15%; desktop Wikipedia views −9% after AI Overviews, marginal; no dose-response across places; Google queries still growing (Table 1, Figure A1)	Supported in narrow form
Effort, more than trust, drives the fall in clicks (Prop. 10)	Informational queries; skeptical users rarely check sources; no rise in clicks after May 2024 errors (Section 5.6)	Consistent; placebo $p = 0.40$–0.70
AI substitution news lowers the value of search-dependent firms only (Prop. 4)	Alphabet CAR −7.0% and −12.6%; Meta flat; pooled $Z = -4.32$ (Table 2)	Supported
Ads in own AI answers raise incumbent value (Props. 1–2)	+6.0% for AI Mode ads; no reaction for AI Overviews ads; pooled $Z = 0.88$ (Table 2)	Partly supported
A rival's ads raise incumbent value when $\eta = 0$ (Prop. 3); little net effect when $\varepsilon w_I \approx 1$ (Prop. 6)	No significant reaction to ChatGPT ads (Table 2); $\varepsilon \approx 1$ (Table 4)	Prop. 3 not supported; Prop. 6 consistent
Scarcity dividend: ad price rises when inventory grows more slowly (Prop. 7)	CPC growth falls one-for-one with click growth, 2023–2026 (Table 4)	Supported, not causal
Search revenue falls behind the ad cycle as AI diffuses (Prop. 4)	Slower growth since 2023, but placebo breaks fit equally well (Table 3)	Inconclusive
Summaries before full answers; little legacy profit, straight to full answers (Prop. 8)	Google: AI Overviews before AI Mode; Bing: full answers in Feb. 2023, share gains stalled after Google's summaries (Section 5.5)	Consistent, not causal
More legacy profit, later full answers (Props. 8–9)	No relation across 15–17 engines (Table A8); higher-value Google markets not delayed	Not supported
Capability and scale speed	In-house models about 12 months earlier ($p =$	Partly supported;

Prediction (source)	Evidence	Verdict
adoption, regulation delays it (Prop. 9)	0.11); larger markets earlier; EU/EEA about 2 and France and Türkiye 7–8 months later	regulation supported

Why Proposition 3 fails. The base model lets a rival's ads affect the incumbent only through consumers. With an advertiser side, the rival's ads also lower the price of the incumbent's ads, and Proposition 6 shows that the net effect is positive only if $\varepsilon w_I < 1$. Table 4 implies ε close to 1, and Google holds most search ad inventory, so the two effects roughly cancel, which matches the null result.

Remaining tests. Three designs could identify the AI effect more cleanly than firm-level time series: country-level search ad spend around the staggered rollouts of AI Overviews and AI Mode, a comparison of high-profit commercial categories with informational categories (Proposition 4), and a panel of AI answers coding sponsored content per answer (Proposition 2); see Online Appendix E. Replication files are described in Online Appendix B.

6. Discussion

6.1 Managerial and Policy Implications

For incumbent search platforms, the relevant question is not whether to place advertising in AI answers but how to clear the recovery threshold in Proposition 4. Investments that make ads more productive in a conversational setting, such as formats that draw on intent expressed over several turns, are of first-order importance, whereas investments that lower inference cost do not raise profit in a competitive equilibrium, because their benefit is passed on to users as lighter ad loads. For AI-native entrants, staying ad-free is an expensive and temporary way to stand apart (Proposition 3). For all firms, the timing of an AI rollout appears to be set less by what a firm

stands to lose than by whether it has its own model and how large its market is (Section 5.5), so waiting for the legacy business to shrink is risky. For advertisers, clicks will remain scarce because users avoid the effort of reading results, and outcome-based pricing is likely to become more attractive.

Advertising inside generated answers raises concerns that a separate column of sponsored links does not: if commercial content shapes the substance of an answer, users may not be able to tell advice from advertising. Disclosure rules that keep ads visibly apart from the answer raise the effective nuisance γ in the short run, but by protecting trust they may preserve differentiation t in the long run; by Proposition 4 they also raise the recovery threshold, which regulators should weigh against the gain in consumer protection. If users rarely verify AI answers (Section 4.8), they cannot discipline a platform that blends ads into them, so disclosure rules matter more when users are complacent. Regulation also shapes how fast AI answers arrive (Section 5.5), and antitrust analysis may need to adapt if competition moves from the search box to the AI answer.

6.2 Limitations and Future Research

The model is deliberately stylized. The core model is static, and a dynamic model in which today's ad load erodes tomorrow's demand would probably yield lower ad intensity. We set subscriptions aside, although a menu combining a free tier with ads and a paid tier without them (Prasad et al. 2003) would weakly dominate either pure model. The share s of AI-amenable queries is exogenous, the advertiser side is reduced-form, and we leave out publishers. The empirical analysis also has limits. The monetization tests rely on a single incumbent, a small number of events, and a benchmark firm (Meta) that faces its own shocks. The timing analyses have between 17 and 47 observations and can detect only large effects, so the capability result (p

= 0.11) is suggestive. Our measures of clicks come from Wikipedia, one destination among many, and the trust-shock test exploits a single episode, so a null result cannot rule out that trust matters elsewhere; a controlled experiment that varies the cost of reading results while holding accuracy fixed would be a sharper test. Several launch dates also come from trade-press reports rather than company documents.

7. Conclusion

Taken together, our results suggest that generative AI has not, in the aggregate, made people search less; it has made them click less. Individual chatbot adopters do run fewer searches (Padilla et al. 2026), but we find no sign that AI adoption reduced search activity across U.S. states or European countries, while the staggered launch of AI Overviews points to fewer clicks out of search. Because sponsored search is sold by the click, the threat is to revenue per search rather than to search itself.

The model explains why advertising inside AI answers is close to unavoidable, since every answer costs money to produce, and shows how that cost is divided between users and advertisers. One of its predictions fails: incumbents with more to lose do not wait longer to launch full answers. Across search engines and across Google's own markets, the pace is set by capability, market size, and regulation. Nor does trust seem to be the main reason users stop clicking: when AI Overviews made widely reported errors in May 2024, clicks did not recover, which is more consistent with users avoiding the effort of reading results.

The October 2026 repricing of Alphabet fits a market coming to terms with answers that are being deployed quickly while their monetization lags behind. Whether search platforms decline

or reinvent themselves will depend less on whether people keep searching than on whether answers can carry advertising without losing the convenience, and the trust, that make them worth using.

Disclosures

Use of generative AI. AI tools (Claude from Anthropic), in line with the INFORMS policy on AI tools, were used for coding and analyses, and editing the manuscript. The author confirms that the author reviewed and takes full responsibility for all content.

Data and code availability. All data used in this paper are publicly available: Wikimedia projectviews dumps, Google Trends, Eurostat (isoc_ai_iaiu, isoc_ci_ifp_fu), World Bank World Development Indicators, StatCounter Global Stats, and company filings on SEC EDGAR. Code and processed data will be deposited in a public repository upon acceptance.

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